


CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

Test Pattern

JEE(Advanced)
FULL SYLLABUS
15-05-2024
JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY
GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

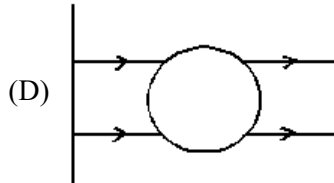
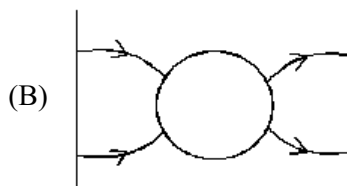
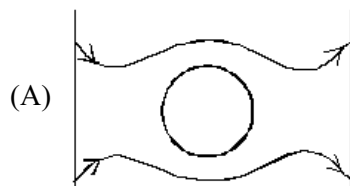
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

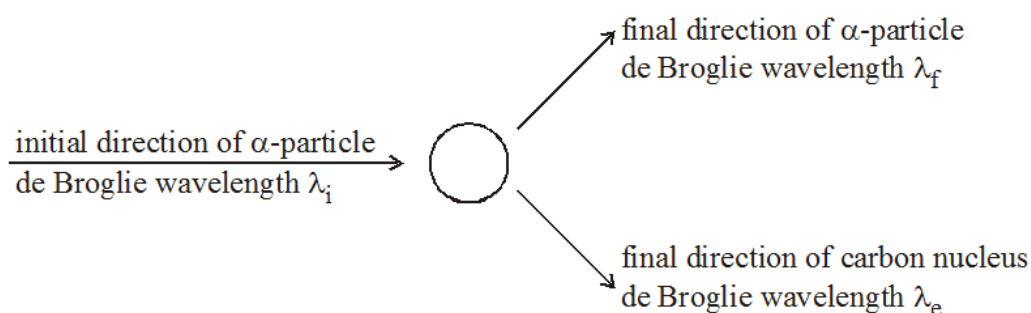
1. On the superposition of two harmonic oscillations represented by $x_1 = a \sin(\omega t + \phi_1)$ and $x_2 = a \sin(\omega t + \phi_2)$ a resulting oscillation with the same time period and amplitude is obtained. The value of $\phi_1 - \phi_2$ is :-
 (A) 120°
 (B) 90°
 (C) 60°
 (D) 15°
2. An uncharged sphere of metal is placed inside a charged parallel plate capacitor. The electric lines of force will look like



3. An electromagnetic wave travelling along z-axis is given as: $\vec{E} = \vec{E}_0 \cos(kz - \omega t)$. Choose the **INCORRECT** options from the following;

- (A) The associated magnetic field is given as $\vec{B} = \frac{1}{c} \hat{k} \times \vec{E} = \frac{1}{\omega} (\hat{k} \times \vec{E})$
- (B) The electromagnetic field can be written in terms of the associated magnetic field as $\vec{E} = c (\vec{B} \times \hat{k})$
- (C) $\hat{k} \cdot \vec{E} = 0, \hat{k} \cdot \vec{B} = 0$
- (D) $\hat{k} \times \vec{E} = 0, \hat{k} \times \vec{B} = 0$

4. An α -particle having a de Broglie wavelength λ_i collides with a stationary carbon nucleus. The α -particle moves off in a different direction as shown below.



After the collision, the de Broglie wavelength of the α -particle and the carbon nucleus are λ_f and λ_e respectively. Which of the following relations about de Broglie wavelength is correct ?

- (A) $\lambda_i < \lambda_f$
- (B) $\lambda_i > \lambda_f$
- (C) $\lambda_f = \lambda_e$
- (D) $\lambda_i = \lambda_e$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

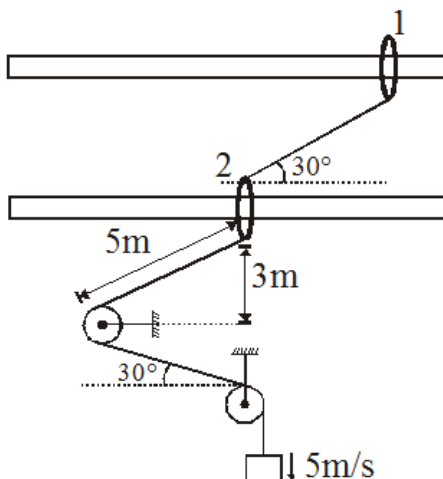
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

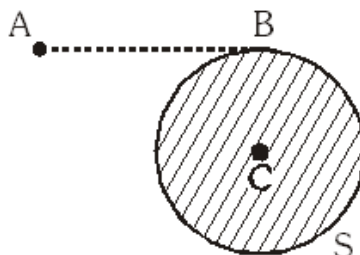
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Consider the following constraint motion. If all the strings remain tight then choose the correct statement(s)

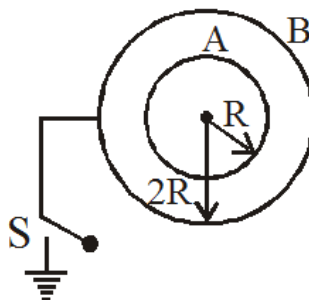


- (A) Velocity of ring (1) is $\frac{25}{4}$ m/s towards left.
- (B) Velocity of ring (1) is $\frac{25\sqrt{3}}{8}$ m/s towards left.
- (C) Velocity of ring (2) is $\frac{25}{4}$ m/s towards left.
- (D) Velocity of ring (2) is 4 m/s towards left.

6. S is a solid neutral conducting sphere. A point charge q of $1 \times 10^{-6} \text{C}$ is placed at point A. C is the centre of sphere and AB is a tangent. $BC = 3 \text{m}$ and $AB = 4 \text{m}$.

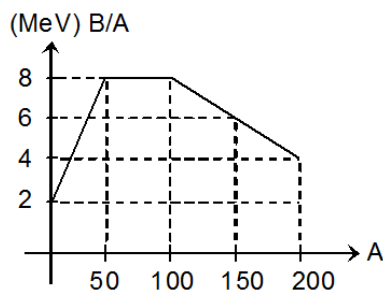


- (A) The electric potential of the conductor is 1.8 kV.
 (B) The electric potential of the conductor is 2.25 kV.
 (C) The electric potential at B due to induced charges on the sphere is -0.45 kV .
 (D) The electric potential at B due to induced charges on the sphere is 0.45 kV .
7. Two concentric conducting spherical shells of radii R and $2R$ having charges q_A and q_B and potential $2v$ and $3v/2$ respectively. Now shell B is earthen by closing switch S. Let charges on spherical shells A & B become q'_A & q'_B respectively, then



- (A) $q_A/q_B = 1/2$ (magnitude)
 (B) $q'_A/q'_B = 1$ (magnitude)
 (C) potential of A after earthing becomes $\frac{3v}{2}$
 (D) potential difference between A and B after earthing becomes $\frac{v}{2}$

8. If a graph of Binding energy (B/A) per nucleon versus mass number (A) looks like as shown in figure. Then using the curve choose **CORRECT** option(s).



- (A) Fusion of two nuclei with mass numbers 30 and 45 will have Q value 99 MeV
- (B) Fission of nuclei with mass number 80 into two nuclei of equal mass number will have Q value 256 MeV
- (C) Fission of nuclei with mass number 150 into two nuclei of equal mass number will release energy
- (D) Fission of nuclei with mass number 80 into two nuclei of equal mass number will release energy
9. An inductor of inductance $L = 10\text{H}$ carrying initial current 5A is connected to a resistor of resistance 20Ω and mass 2kg and specific heat $0.5\text{Jkg}^{-1}\text{K}^{-1}$. Assume that all energy given to resistor is used to increase its temperature .
- (A) The total change in temperature of resistor will be 125°C .
- (B) The total change in temperature of resistor will be 250°C .
- (C) The total charge passed through the resistor is 2.5 C .
- (D) The total charge passed through the resistor is infinite.
10. A ball of density ρ is dropped onto a horizontal solid surface. It bounces elastically from the surface and returns to its original position in time t_1 . Next, the ball is released from the same height, but this time it strikes the surface of the liquid of density $\rho_L (> \rho \text{ but less than } 2\rho)$, and takes t_2 second to come back to its original height:
- (A) $t_2 > t_1$
- (B) $t_2 < t_1$
- (C) The motion of the ball is not simple harmonic
- (D) If $\rho = \rho_L$, then the speed of the ball inside the liquid will be independent of its depth. (neglect viscous force)

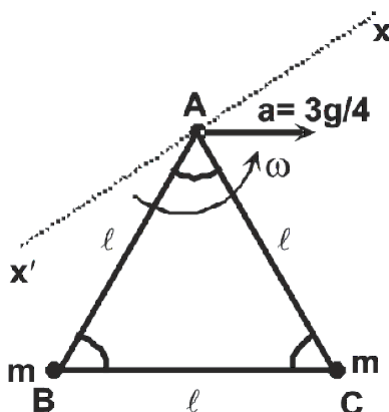
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

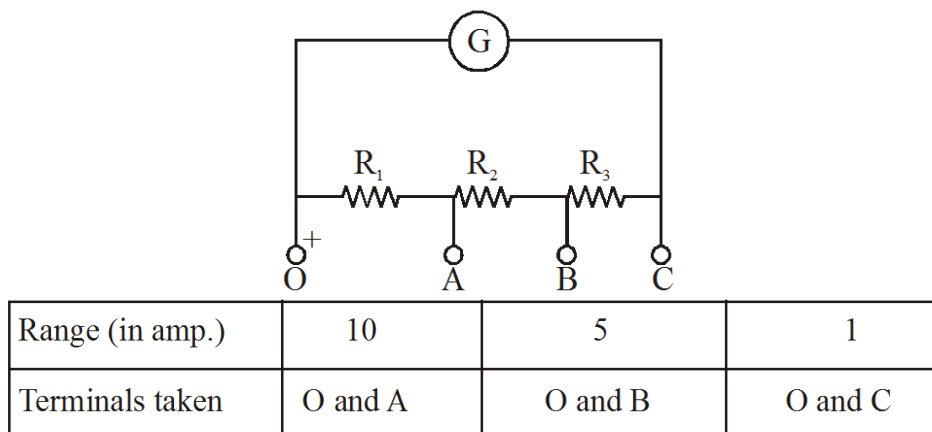
Zero Marks : 0 In all other cases.

1. A rigid triangular frame consists of three massless rods of length $\ell = 2.5$ m each and point masses of mass m each at vertex B and C respectively. Frame is hanging vertically from point A about which it can rotate freely about an axis xx' which is perpendicular to plane of frame as shown in figure. Point of suspension of frame, i.e. A, is accelerating with constant acceleration $a = 3g/4$ in horizontal direction and initially frame is at rest w.r.t. support A. Minimum initial angular velocity $\omega = x(3)^{\frac{1}{4}}$ (in rad/s) provided to system, so that it can complete vertical circular motion in the frame of support A. Calculate the value of x . (take $g = 10 \text{ m/s}^2$)



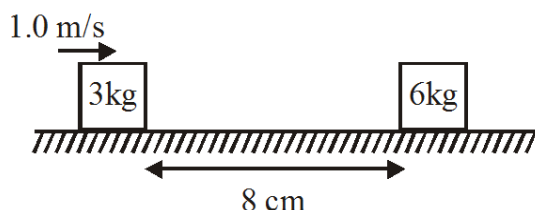
2. For an equilateral prism, it is observed that when a ray strikes grazingly at one face it emerges grazingly at the other. Find the refractive index of the prism.

3. The following figure shows a circuit diagram of an ammeter of three different ranges. The specification of the ammeter is

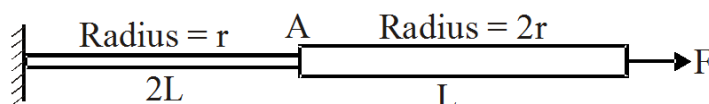


The resistance of the galvanometer coil is 99Ω . The range of the galvanometer is 10 mA. Find the $\left(\frac{R_3}{R_1 + R_2} \right)$.

4. A monochromatic point source of light is placed at a distance 'd' from a small flat metal surface. Photo-electrons are ejected at rate n. If the source is brought nearer to distance d/2, the rate becomes p times of its initial value. Then find the value of p.
5. A tuning fork P of unknown frequency gives 7 beats in 2 seconds with another tuning fork Q. When Q is moved towards a wall with a speed of 5 m/s, it gives 5 beats per second for a stationary observer located left to it. On filing, P gives 6 beats per second with Q. What is the frequency (in Hz) of P? Assume speed of sound = 332 m/s.
6. The friction coefficient between the horizontal surface and each of the block shown in the figure is 0.4. The collision between the blocks is perfectly elastic. If the separation (in cm) between them when they come to rest is x then find the value 8x.



7. A uniform rod of length 0.3m, hanging vertically. The free end acquires velocity of v m/s, when it is hit at its free end. Find the maximum value of v in m/s for which the rod does not reach horizontal position.
8. Two steel wires of radii r and 2r are connected together end to end and tied to a wall as shown.



The force stretches the combination by $\frac{27}{4}$ mm. How far does the junction point A move. (in mm)

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

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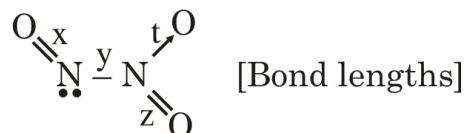
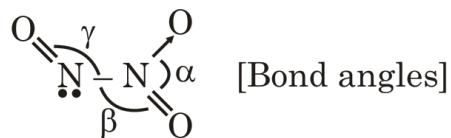
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Which is NOT a redox reaction ? (ignore balancing)

- (A) $\text{Pb}_3\text{O}_4 + \text{HNO}_3 \rightarrow$
 (B) $\text{CaOCl}_2 + \text{dil. HCl} \rightarrow$
 (C) $\text{XeO}_6^{4-} + \text{F}^{-1} + \text{H}^+ \rightarrow$
 (D) $\text{Na}_2\text{S} + \text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}] \rightarrow$

2. Consider following structure of N_2O_3 :



Choose incorrect option :

I. $y > z = t > x$

II. $\alpha > \beta > \gamma$

(A) I only

(B) II only

(C) Both

(D) None

3. At low pressure, if $RT = 2\sqrt{P \times a}$, then the molar volume occupied by a real gas is, where 'a' is vanderwaal's coefficient.

- (A) $\frac{2RT}{P}$ (B) $\frac{2P}{RT}$
(C) $\frac{RT}{2P}$ (D) $\frac{3RT}{P}$

4. The rate constants for the bromination of several disubstituted stilbenes are given. Find the incorrect relation between rate constants for different X and Y?

The rate the table



X	Y	$K(\times 10^3 \text{ l/mol-min})$
OMe	Me	K_1
OMe	Cl	K_2
OMe	OMe	K_3
OMe	NO_2	K_4

- (A) $K_3 > K_1 > K_2$
(B) $K_1 > K_2 > K_4$
(C) $K_3 > K_2 > K_4$
(D) $K_1 > K_3 > K_4$

SECTION-I (ii) : (Maximum Marks: 24)

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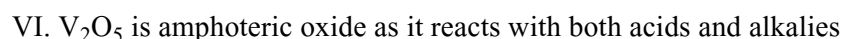
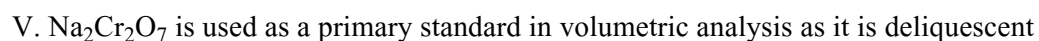
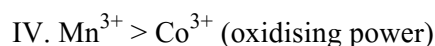
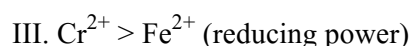
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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Choose correct reaction(s)/order(s)/statement(s)



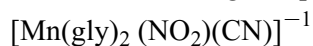
(A) I, III

(B) IV, V

(C) IV, VI

(D) I, II, V, VI

6. Consider following complex :



Choose incorrect statements regarding this complex :

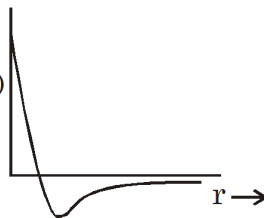
- (A) There are 4 geometrical isomers in which NO_2 and CN are in cis position w.r.t. each other
- (B) Splitting energy is less than pairing energy
- (C) $\text{CFSE} = -0.6\Delta_0$
- (D) $\text{Mn} - \text{C}$ bond will have partial double bond character
7. 100 ml of 1 M HCN is titrated with 5 M KMnO_4 in acidic medium, A gas (B) is liberated which turns lime water milky, along with an anion which gives brown ring test (central atom in the highest oxidation state). When this gas is passed through lime water, 10 gms of a white ppt. is obtained. Identify the correct statement(s) :-
- (A) Normality of the HCN solution would be 10N
- (B) Volume of the KMnO_4 solution at equivalence point would be 40 ml
- (C) Equivalent mass of HCN in this process is greater than its molecular mass
- (D) White precipitate formed in this reaction would be CaC_2

8. Choose the correct statement(s) among the following regarding hydrogen atom?

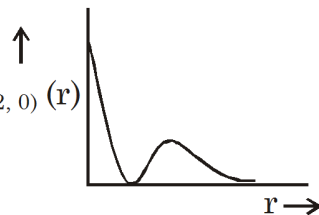
(All graphs given below are indicative in nature, not to scale)

- (A) For an atomic orbital, radial wave function $R(r)$ depends on principle, azimuthal, and magnetic quantum number.

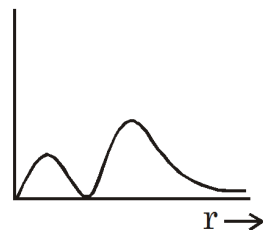
- (B) Plot of $R_{(2,0)}(r)$ versus r (distance from nucleus), for 2s electron is $R_{(2,0)}(r)$



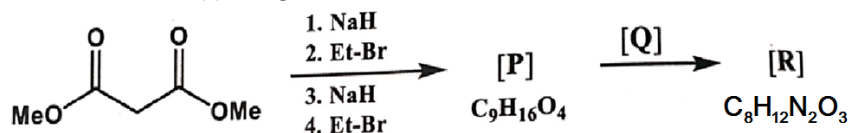
- (C) Plot of $R_{(2,0)}^2(r)$ versus r (distance from nucleus), for 2s electron is $R_{(2,0)}^2(r)$



- (D) Plot of $4\pi r^2 R_{(2,0)}^2(r)$ against r (distance from nucleus) is $4\pi r^2 R_{(2,0)}^2(r)$

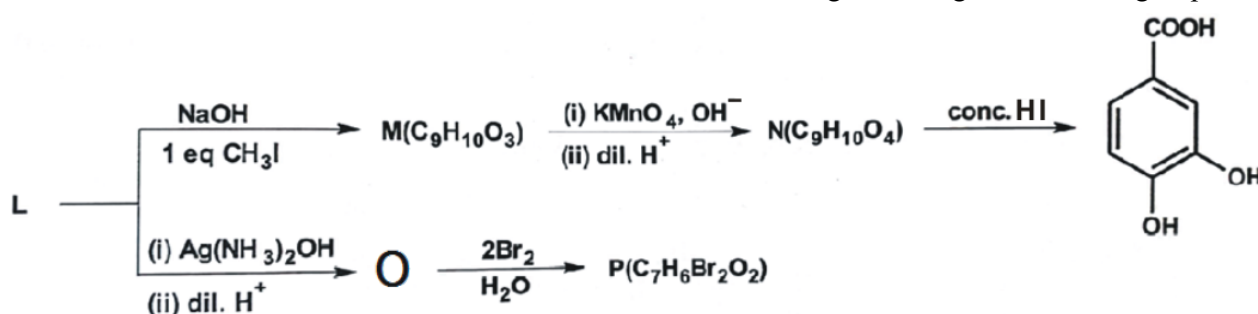


9. Choose the **CORRECT** statement(s) for given reaction scheme



In the above reaction scheme, if [Q] is urea and [R] is veronal, then correct statement(s) is/are :-

- (A) [R] is a derivative of barbituric acid
 (B) [R] is used as a hypnotic.
 (C) [R] contains two delocalized lone pair.
 (D) There will be three C=O groups in [R]
10. Vanillin (L) $\text{C}_8\text{H}_8\text{O}_3$ is isolated from vanilla beans. It gives an intense colour with FeCl_3 and a positive Tollen's test. It is not steam-distilled and does not react with HCl . It goes through the following steps :



Among the following pair(s) which can be distinguished by aq. NaHCO_3 is/are :-

- (A) M, N (B) N, P
 (C) L, O (D) P, O

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
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Zero Marks : 0 In all other cases.

- Consider following compounds :
 MnO_2 , HgI_2 , $\text{Cu}(\text{CN})_2$, $\text{Cu}_2[\text{Fe}(\text{CN})_6]$, $\text{Fe}(\text{CN})_3$, $\text{HgSO}_4 \cdot 2\text{HgO}$, Cu_2O , MnS , $[\text{Ni}(\text{CN})_4]^{2-}$
 Let,
 x = number of yellow coloured compounds
 y = number of brown coloured compounds
 Find $\frac{xy}{2}$
- Consider following molecules/ions :
 SF_4 , ClF_3 , N_2O , XeO_6^{4-} , Al_2Cl_6 , BrF_5 , C_6H_6
 Let
 x = number of molecules/ion having all bond length equal
 y = number of polar as well as planar species. Find $\frac{y-x}{3}$
- Solid AgNO_3 is added to an aqueous solution containing 0.2 M $\text{Br}^\ominus$ and 0.1 M $\text{I}^\ominus$. Assuming that AgNO_3 addition does not change volume of this solution, calculate % $[\text{I}^\ominus]$ precipitated when $[\text{Br}^\ominus]$ just starts precipitating : $K_{\text{sp}}(\text{AgBr}) = 10^{-13} \text{ M}^2$; $k_{\text{sp}}(\text{AgI}) = 10^{-17} \text{ M}^2$
- An ideal gas having initial pressure P , volume V and temperature T is allowed to expand adiabatically and reversibly until its volume becomes $5.66V$ while its temperature falls to half of initial temperature. The work done by the gas is represented by x times PV . Find the value of x .
 $(10^{0.3} = 2, 10^{0.48} = 3, 10^{0.75} = 5.66)$

5. X constitutes an FCC lattice with Y occupying all octahedral voids and Z occupying all tetrahedral voids. One such isolated unit cell is divided into 8 equal parts, then select total number of correct statements among the following
- Every equal part contains all X, Y, and Z.
 - In smaller part, 4 corners are occupied by X and 4 corners are occupied by Y.
 - In smaller part, 4 corners are occupied by X, 2 by Y and 2 by Z.
 - In smaller part, effective numbers are $X = \frac{1}{2}$, $Y = \frac{1}{2}$, $Z = 1$
 - Co-ordination number of X with respect of Z in the lattice is 8
 - Co-ordination number of X with respect of Y in the lattice is 6
 - Number of next nearest neighbour of Y is 8
 - If edge length of $\frac{1}{8}$ -th equal part of the isolated unit cell is b, then the minimum distance between two Y atoms would be $\sqrt{2}b$
6. An alkene C_nH_{2n} is converted by a reduction procedure to alkane C_nH_{2n+2} . The % change in the molecular weight = 3.57. What is the value of n. (rounded off to the nearest integer.)
7. (a) Leucine , Lysine , Serine, Cysteine , Proline, Threonine, Arginine , Histidine
Number of Essential amino acids in above mentioned are X.
(b) In phosphodiester linkage of a Dinucleotide, carbon number (Y) of one sugar and carbon number (Z) of other sugar are joined by phosphate. Then $X \div (Y + Z)$ is :
8. The amount of product produced in the following reactions is _____ kg.

$$C_2H_4(2\text{kg}) + H_2(2\text{kg}) \xrightarrow{\text{Wilkinson's Catalyst}} \text{product (90 \% catalytic conversion)}$$

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. $\lim_{x \rightarrow 0} \frac{\ln(\sec(x) \sec(e^2x) \dots \sec(e^{50}x))}{e^2 - e^{2 \cos x}}$ is equal to
- (A) $\frac{e^{100} - 1}{2(e^2 - 1)}$
- (B) $\frac{e^{100} - 1}{(e^2 - 1)}$
- (C) $\frac{e^{100} - 1}{2e(e^2 - 1)}$
- (D) $\frac{e^{50} - 1}{2e(e^2 - 1)}$
2. Let a be a fixed real number. Consider the equation,
 $(x + 2)^2(x + 7)^2 + a = 0, x \in \mathbb{R}$
This equation has exactly one double root. Possible value of a is
- (A) 0
- (B) $-\frac{9}{2}$
- (C) $-\left(\frac{5}{2}\right)^4$
- (D) -2

3. If the equation $x^3 - 3x + 1 = 0$ has three real roots x_1, x_2, x_3 where $x_1 < x_2 < x_3$ then the value of $(\{x_1\} + \{x_2\} + \{x_3\})$ is, (where $\{\cdot\}$ is fractional part function)
- (A) $\frac{3}{2}$
(B) 1
(C) 2
(D) $\frac{5}{2}$
4. Equation of a plane passing through the line of intersection of the planes $x + 2y + z = 10$ and $3x + y - z = 5$ and passing through origin is
- (A) $5x + 3z = 0$
(B) $5x - 3z = 0$
(C) $5x + 4y + 3z = 0$
(D) $5x - 4y + 3z = 0$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Given, $f(x) - \mu \int_0^{\pi/4} \sin^2 x \cos^{-5} x \cos t f(t) dt = \sin^2 x \cos^{-5} x$ (where μ is real constant) then

(A) $f(x) = \frac{3}{3 - \mu} \sin^2 x \cos^{-5} x$

(B) $f(x) = \frac{3}{\mu} \sin x \cos x$

(C) $f(x)$ is periodic with period π

(D) $f(x)$ is periodic with period 2π

6. A differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies following relations

(i) $f(x) - f(y) = (x - y)f' \left(\frac{x + y}{2} \right), \forall x, y \in \mathbb{R}$

(ii) $f(0) = f(1) = 0, f(2) > 0$

Which of the following is/are true

- (A) $f(x)$ is an onto function
- (B) $f(x)$ has an extrema at $x = \frac{1}{2}$
- (C) $f(x)$ is a many one function
- (D) $\lim_{x \rightarrow -\infty} f(x) = -\infty$
7. Let $\hat{a}$ and $\hat{c}$ be unit vectors and $|\vec{b}| = 4$. If the angle between $\hat{a}$ and $\hat{c}$ is $\cos^{-1} \left(\frac{1}{4} \right)$ and $\vec{b} - 2\hat{c} = \lambda\hat{a}$ then the value of λ can be :
- (A) 2
- (B) -3
- (C) 3
- (D) -4
8. Which of the following is/are true ?
- (A) If z is a complex number satisfying $|z - 2 - i| = |z| \left| \sin \left(\frac{\pi}{4} - \arg z \right) \right|$ then locus of z is a parabola
- (B) If z is a complex number satisfying $|z - \cos^{-1}(\cos 2)| - |z - \sin^{-1}(\sin 12)| = 8(\pi - 3)$ then locus of z is a hyperbola
- (C) If z is a complex number satisfying $z^2 + az + a^2 = 0$ then $|\arg z|$ can be $\frac{2\pi}{3}$ ($a > 0$)
- (D) If z is a complex number satisfying $z^2 + az + a^2 = 0$ then $\arg(z) = \frac{\pi}{3}$ ($a \in \mathbb{R}$)

9. The differential equation $(xy^3(1 + \cos x) - y) dx + xdy = 0$ represents the curve

$\frac{x^2}{2y^2} = \frac{x^3}{b} + x^2 \sin x + cx \cos x - d \sin x + k$ then, (where $b, c, d \in \mathbb{Z}$ ($\mathbb{Z}$ is set of integers & k is constant of integration))

(A) $d = 2$

(B) $c = 3$

(C) $b = 3$

(D) $c = 2$

10. If $\int_0^{\pi} (\sin^3 x) e^{-\sin^2 x} dx = \alpha - \frac{\beta}{e} \int_0^1 \sqrt{t} e^t dt$ then

(A) $\alpha = 2$

(B) $\beta = -3$

(C) $\alpha + \beta = 5$

(D) $\alpha - \beta = 5$

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $f(x) = \left([\{x\}] \tan^{-1} \left(\frac{x^2 - 3x - 1}{x^2 - 3x + 5} \right) + 3 - x^7 \right)^{\frac{1}{7}}$
Where $[k]$ and $\{k\}$ denotes greatest integer and functional part of k respectively, then the value of $f^{-1}(50) - f(50) + f(f(100))$, is
2. The number of 5 digit numbers that can be formed using the digits 1, 2, 3, 4, 5, 6, 7, 8, 9 in which one digit appears once and two digits appear twice (eg., 44533) is k , then the value of $\frac{k}{1890}$ is equal to
3. If sum of the values of x satisfying the equation $\sqrt{2} \sin \left(x + \frac{\pi}{4} \right) (\sec^2 x - (1 + \sqrt{3}) \tan x + 2 + \sqrt{3}) = \sqrt{9 + 18 \sin x \cos x}$
in $[0, 2\pi]$ is $\left(\frac{p}{q} \right) \pi$, where p and q are co-prime, then $|p - q|$ is equal to
4. If the least positive value of x satisfying the equation $\frac{\sin x}{\cos 3x} + \frac{\sin 3x}{\cos 9x} + \frac{\sin 9x}{\cos 27x} = 0$ is $\frac{\pi}{k}$ where $k \in \mathbb{N}$
then $\left(\frac{k}{5} \right)$ is equal to
5. If $\frac{\cos x + \cos y + \cos z}{\cos(x + y + z)} = 2$ and $\frac{\sin x + \sin y + \sin z}{\sin(x + y + z)} = 2$
then the value of $\cos(x + y) + \cos(y + z) + \cos(z + x)$ is equal to :
(where $x, y, z \in \mathbb{R}$)
6. Let $(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r$ where $a_0, a_1, a_2, \dots, a_{20}$ are constant then the value of $\frac{a_7}{a_{13}}$ is equal to
7. The greatest possible integral value of $\frac{b - a}{\tan^{-1} b - \tan^{-1} a}$, (where $0 < a < b < \sqrt{3}$) is
8. If $\int_0^{\pi/2} \frac{dx}{(\sqrt{\cos x} + \sqrt{\sin x})^4} = J$ then $\frac{1}{J}$ is

Space for Rough Work

NAME OF THE CANDIDATE

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and shall abide by them.

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I have verified the identity, name and Form
number of the candidate, and that question
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